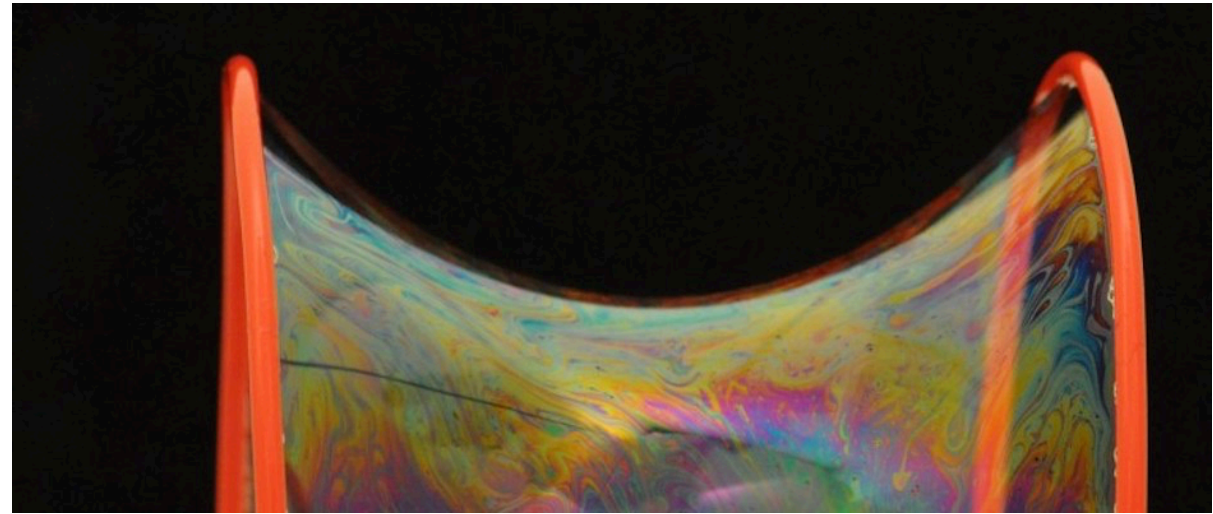


Day 29 - Calculus of Variations



Announcements

- Midterm 2 is posted (Due 10 April 2026 at 11:59 pm)
- **Office Hours this week:** Midterm 2 Help

Calculus of Variations

Variational calculus is a mathematical method to find functions that optimize a certain quantity. We will use variational calculus to derive the **Euler-Lagrange equations** for a set of **generalized coordinates** (i.e., $q, \dot{q}$). This is fundamental to Lagrangian mechanics.

$$L(q, \dot{q}, t) = T(\dot{q}) - U(q)$$
$$S = \int_{t_1}^{t_2} L(q, \dot{q}, t) dt \quad \delta S = 0$$
$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) - \frac{\partial L}{\partial q} = 0$$

Clicker Question 29-1

The generic segment, ds , of a curve in 2D Cartesian coordinates is given by

$$ds = \sqrt{(dx)^2 + (dy)^2}$$

The integral of ds from s_1 to s_2 gives the length of the curve, l . What is the correct expression for l ?

1. $l = \int_{s_1}^{s_2} ds$

2. $l = \int_{s_1}^{s_2} \sqrt{(dx)^2 + (dy)^2}$

3. $l = \int_{s_1}^{s_2} \sqrt{1 + (dy/dx)^2} dx$

4. $l = \int_{s_1}^{s_2} \sqrt{(dx/dy)^2 + 1} dy$

5. More than one of the above

Clicker Question 29-2

I can explain why:

$$\int_{s_1}^{s_2} f((Y(x), Y'(x), x) dx > \int_{s_1}^{s_2} f((y(x), y'(x), x) dx$$

where $Y(x) = y(x) + \alpha\eta(x)$, the true path plus an error term.

1. Yes, I can explain why
2. I think I can explain why
3. I'm having trouble seeing why
4. I don't think I can explain why

Clicker Question 29-3

For the function $Y(x) = y(x) + \alpha\eta(x)$, where $y(x)$ is the true path, $\eta(x)$ is a small error term, and α is a small parameter, what is the derivative of $Y(x)$ with respect to α ?

$$\frac{dY}{d\alpha} = ?$$

1. $y(x)$
2. $\eta(x)$
3. $\eta'(x)$
4. $\alpha\eta(x)$
5. $y'(x) + \alpha\eta'(x)$

Clicker Question 29-4

For the function $Y'(x) = y'(x) + \alpha\eta'(x)$, what is the derivative of $Y'(x)$ with respect to α ?

$$\frac{dY'}{d\alpha} = ?$$

1. $y'(x)$
2. $\eta'(x)$
3. $\eta''(x)$
4. $\alpha\eta'(x)$
5. $y''(x) + \alpha\eta''(x)$

Clicker Question 29-5

The "surface term" that we computed for $\int_{s_1}^{s_2} \eta'(x) \frac{df}{dy'} dx$ is:

$$\left[\eta(x) \frac{df}{dy'} \right]_{x_1}^{x_2} = 0$$

I can explain why this surface term is equal to zero:

1. Yes, I can explain why
2. I think I can explain why
3. I'm having trouble seeing why
4. I don't think I can explain why
5. I don't know what a surface term is

Clicker Question 29-6

We completed this derivation with the following mathematical statement:

$$\int_{s_1}^{s_2} \eta(x) \left[\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) \right] = 0$$

where $\eta(x)$ is an arbitrary function. What does this imply about the term in square brackets?

1. The term in square brackets must be a pure function of x .
2. The term in square brackets must be a pure function of y .
3. The term in square brackets must be a pure function of y' .
4. The term in square brackets must be zero.
5. The term in square brackets must be a non-zero constant.